

## Senior AI / Machine Learning Engineer | LLM Agents, Forecasting & Production ML

Senior AI / ML Engineer, 6+ years on production ML over 10M+ record datasets, now focused on **LLM agent systems**: multi-agent RAG, tool-using diagnostic agents and LLM orchestration inside enterprise platforms, owned from business framing to industrialization, leading teams of up to 5 and reporting to C-level. Computational Biology background (*Genome Biology*): strong on statistical modeling and rigorous evaluation.

## Experience

### Enel, Senior AI Engineer

Florence, IT

June 2023 – present

- **Enel-wide incident management platform.** Took ownership and, by designing and shipping new features, grew its coverage to **~90% of Enel's IT incidents**, with **~5% now closing end-to-end with no human involved**.
- **AI-based incident resolution for IT&S — built from scratch, Italy & Spain, in production.** **Owned architecture and development decisions**, with ops teams in Rome and Seville, in **Italian, Spanish and English**. *Italy (fibre)*: a tool-using agent executing NOC runbooks step by step — queries SolarWinds, interrogates Huawei/Nokia devices over CLI, reasons over each result and writes its diagnosis into the ServiceNow ticket. *Spain (Seville, ~1,000 incidents and ~8,000 device diagnostics/month)*: a **multi-agent RAG platform** — LLM orchestrator, memory service, retrieval over runbooks, connectors to NetCool/SolarWinds/ServiceNow, **Text2SQL** over diagnostic history; output signed off by IT&S business stakeholders. Stack: LangChain, LlamaIndex, AWS Bedrock, FastAPI, Kubernetes, pgvector.
- **VAudit — secure conversational access to audit documentation, in production.** RAG assistant with **per-user authorization**, so answers are grounded only in documents the requester may see, with citations to the source. **Serverless on AWS**, shipped via a governed CI/CD pipeline.
- **Customer-base forecasting programme — zero to production, team of 5, C-level sponsor.** Led a multi-year programme under **CEO-level attention**: hierarchical forecasting of **churn and payment-default disconnections** as a function of price evolution, at both aggregate and **micro-cluster** level, with **conformal prediction** for calibrated uncertainty and **scenario simulation** of customer-base evolution. Owned architecture and delivery, **managed a team of 5**, and ran the relationship with the CB business unit and executive stakeholders. Stack: Statsmodels, skforecast, Nixtla, Spark, FastAPI.
- Led AI PoCs with **AFC, FLEX and Retail**; set group AI governance standards.

### Enel, AI Engineer / Data Scientist

Rome, IT

Sept 2020 – Jun 2023

- Led technical development of **churn and claim prediction** on **18M contracts** (PySpark), **cutting churn by 15%**; owned ETL, features, training and serving. Stack: XGBoost, CatBoost, H2O, MLlib, Dash.
- Delivered **scenario-based churn volume forecasting, customer segmentation** and **cross-selling models for electrification** for the Retail & Market business unit.
- Built a **real-time anomaly detection dashboard for BESS** (battery energy storage), deployable on edge; plus internal tooling and MLOps standardization (MLflow, Airflow, Luigi).

### DKFZ|EMBL, Computational Biologist

Heidelberg, DE

Sept 2019 – Sept 2020

- Designed a pipeline for **high-dimensional biological data analysis** (single-cell eQTL mapping) using **linear mixed models and neural embeddings**; led to a publication in *Genome Biology* [↗](#).

## Education

**MSc**   **University of Trento**, Computational Biology — GPA 4.0/4.0

Sept 2017 – Sept 2019

## Side Projects

[github.com/gioalvari](https://github.com/gioalvari) [↗](#) — local LLM serving and agent-infrastructure experiments; **global-forecasting** library.

## Skills

**LLM & Agents**: LangChain, LlamaIndex, RAG, multi-agent orchestration, tool/function calling, Text2SQL, AWS Bedrock, pgvector, MongoDB.

**ML & Forecasting**: Python, SQL, scikit-learn, XGBoost, PyTorch, hierarchical forecasting (Nixtla, skforecast), conformal prediction, causality.

**Engineering**: FastAPI, Pydantic, Docker, **Kubernetes**, AWS (Bedrock, serverless), PySpark, MLflow, CI/CD.   **Spoken**: IT, EN (C1), ES (B1).